

MACAE, Brazil

WMO: 868910

Lat: 22.343S

Lon: 41.766W

Elev: 8

StdP: 14.69

Time Zone: -3.00 (SAB)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	56.8	58.8	52.1	57.9	66.1	54.6	63.4	66.4	18.7	75.1	16.9	76.2	3.6	320	0.301

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	17.6	98.3	78.9	95.0	77.9	92.1	77.2	81.7	91.0	80.6	88.7	79.8	87.4	9.0	30

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
79.2	151.6	84.3	78.8	149.7	84.0	77.4	142.6	82.6	45.4	91.6	44.3	89.2	43.4	87.3	88.2

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 18.1	16.1	14.2	51.8	103.8	2.0	3.6	50.4	106.4	49.2	108.5	48.1	110.5	46.7	113.0		
(5)			51.0	84.8	2.0	2.2	49.6	86.4	48.4	87.7	47.2	88.9	45.8	90.5		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	76.6	81.7	82.6	80.8	77.7	73.8	71.9	71.6	72.0	74.3	75.7	77.2	79.9
	DBStd	5.77	4.33	3.52	3.90	3.83	3.84	4.06	4.42	4.49	5.11	4.98	4.90	4.37
	HDD50	0	0	0	0	0	0	0	0	0	0	0	0	0
	HDD65	7	0	0	0	0	0	1	3	2	1	0	0	0
	CDD50	9692	982	914	955	831	737	658	669	681	728	795	815	927
	CDD65	4224	517	494	490	381	272	209	207	218	278	331	365	462
	CDH74	31436	4742	4742	4278	2843	1595	1153	1160	1248	1568	1877	2451	3779
	CDH80	11265	1928	1985	1565	869	419	330	375	437	593	612	793	1359
(14) Wind	WSAvg	6.5	7.1	7.0	5.5	5.5	5.1	5.3	6.1	6.9	7.7	7.5	7.1	6.7
(15) Precipitation	PrecAvg	53.6	7.6	4.8	5.8	4.4	2.5	1.6	1.8	1.7	3.9	4.6	7.3	7.9
	PrecMax	65.5	13.9	15.4	14.8	8.2	3.5	4.1	4.6	3.7	10.3	8.2	12.2	10.2
	PrecMin	40.4	1.3	0.3	1.3	1.9	0.6	0.5	0.4	0.6	0.7	2.1	4.1	5.0
	PrecStd	7.0	3.1	3.4	3.1	2.0	0.8	0.9	1.2	0.9	2.3	1.9	2.1	1.9
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	100.4	101.4	100.3	95.1	91.8	91.6	92.5	96.4	97.0	96.7	97.1	100.7
		MCWB	80.9	80.0	80.7	76.9	74.6	73.3	71.9	72.1	75.3	76.5	78.1	81.0
	2%	DB	95.4	96.5	93.7	90.5	87.9	87.4	87.9	89.3	91.3	90.0	91.7	94.6
		MCWB	79.4	78.7	79.2	77.0	75.1	72.3	71.3	71.1	74.3	75.4	77.5	78.8
	5%	DB	91.8	92.9	90.0	87.6	84.4	83.8	84.3	84.5	86.3	86.2	87.8	90.0
		MCWB	78.2	78.4	78.7	76.9	74.0	71.7	70.1	70.8	72.6	74.2	77.3	77.7
	10%	DB	89.4	89.6	87.6	84.5	81.4	80.5	80.6	80.6	82.4	82.6	84.2	87.2
		MCWB	77.5	77.5	78.5	76.0	73.4	71.1	70.4	70.1	71.2	73.3	75.9	77.3
	0.4%	WB	83.2	82.4	83.2	80.8	79.5	76.1	75.2	75.1	77.4	79.5	80.5	82.1
		MCDB	93.3	94.8	92.7	86.5	83.9	84.4	83.9	86.0	90.4	90.1	89.2	92.0
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	2%	WB	81.4	80.7	81.1	79.5	77.4	74.6	73.6	73.5	75.7	77.4	79.3	80.6
		MCDB	90.4	89.9	89.5	85.9	82.9	82.1	81.9	83.0	86.0	86.2	87.2	88.5
	5%	WB	80.1	79.8	80.1	78.4	76.1	73.3	72.4	72.2	74.4	76.1	78.2	79.3
		MCDB	87.5	88.1	87.7	85.0	81.9	80.0	80.2	80.6	82.8	83.3	84.9	86.7
	10%	WB	79.1	79.0	79.1	77.2	74.6	72.1	71.1	71.1	72.9	74.8	77.1	78.4
		MCDB	85.8	86.5	86.0	83.2	80.3	78.5	78.0	78.5	79.4	80.5	82.4	85.2
(35) Mean Daily Temperature Range	5% DB	MDBR	16.3	17.6	15.6	16.1	16.6	18.5	18.6	18.6	17.1	14.6	14.1	14.8
		MCDBR	21.8	22.2	20.3	19.9	21.4	23.1	24.3	25.9	24.5	21.0	20.3	20.8
		MCWBR	7.5	7.1	8.1	8.6	9.6	10.5	9.5	10.2	8.7	7.9	7.3	7.5
	5% WB	MCDBR	18.6	18.4	17.8	17.5	17.7	20.5	20.9	21.4	20.9	18.0	16.7	17.4
		MCWBR	7.5	7.4	7.9	8.2	9.3	10.7	10.6	10.2	8.4	7.5	7.1	7.4
(40) Clear-Sky Solar Irradiance	taub		0.407	0.399	0.402	0.383	0.383	0.375	0.370	0.408	0.476	0.464	0.438	0.422
	taud		2.480	2.489	2.491	2.529	2.497	2.514	2.509	2.382	2.167	2.257	2.363	2.429
	Ebn at Noon		298	296	287	279	266	261	267	267	261	274	287	294
	Edh at Noon		37	36	35	31	30	28	29	36	48	45	42	39
(44) All-Sky Solar Radiation	RadAvg		1990	2040	1739	1510	1218	1138	1173	1401	1507	1680	1697	1880
	RadStd		297	184	137	89	50	69	84	93	161	184	153	211

Historical Trends

	DBAvg	Heating			Cooling			Degree-Days			
		99% DB		99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (3 neighbors)	N/A	N/A	N/A	-3.04	N/A	-2.52	-2.99	N/A	N/A	N/A	N/A

Nomenclature: See separate page